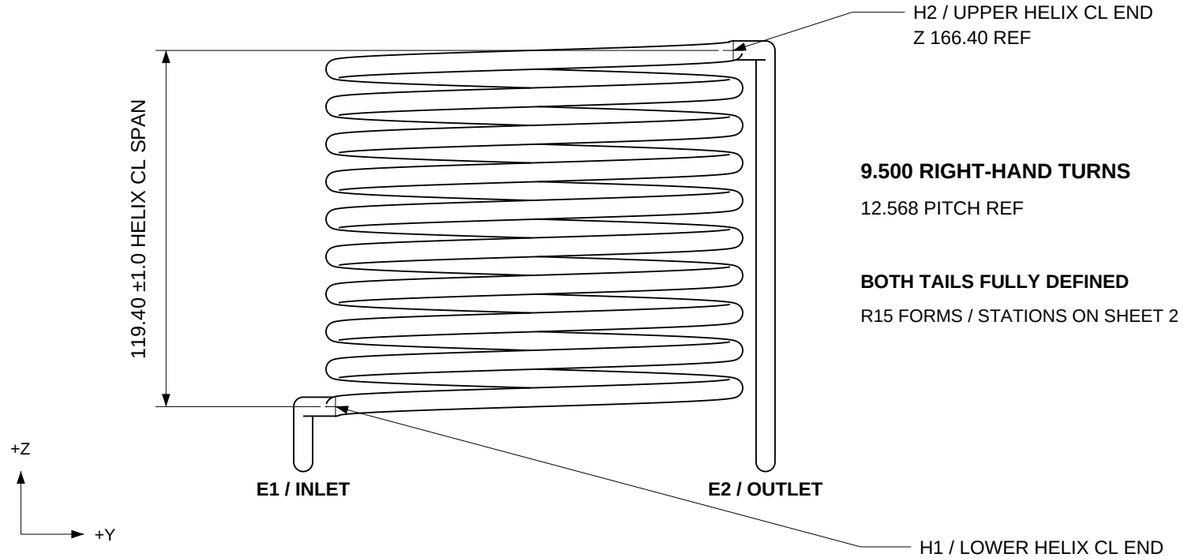
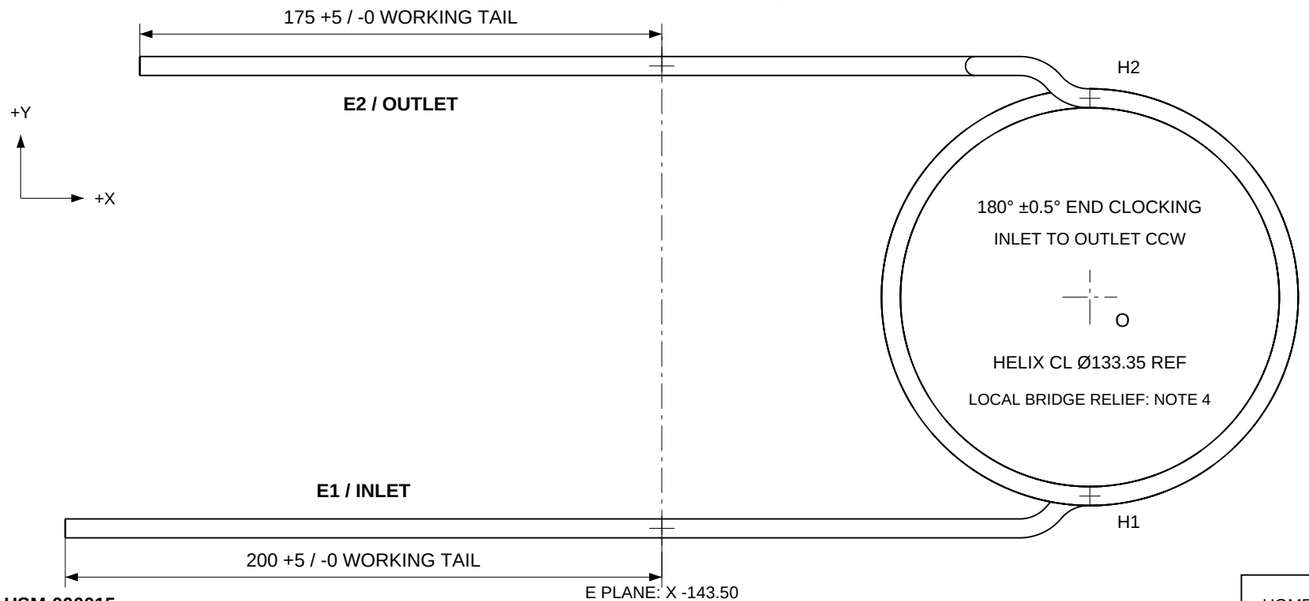
INSTALLED ELEVATION - LOOKING ALONG -X

TUBE OUTLINES - NOMINAL WINDING - SCALE 1:2



PLAN - LOOKING ALONG -Z

TUBE OUTLINES - NOMINAL WINDING - SCALE 1:2



PRODUCTION FORM DEFINITION

ONE CONTINUOUS COPPER TUBE - FINISHED FORM

TUBE STOCK	1/4 IN OD x 0.031 IN WALL ACR COPPER
FINISHED WINDING	9.500 RH TURNS / 180° END CLOCKING
HELIX CL SPAN	119.40 ±1.0
NOMINAL PITCH	12.568 REF
ALL TAIL BENDS	CL R15.0 ±0.5
FORMING DATA	ANGLES / STRAIGHT LENGTHS REF; SHEET 2
EXIT E1 / E2	Y -77.50 / +77.50; Z 28.425; ±0.25
EXIT DIRECTIONS	PARALLEL TO -X WITHIN 1°
STOCK CUT	4811 +10 / -0; FINISH ENDS AFTER FORMING

MANUFACTURING AND ACCEPTANCE

1. Units: mm. O is the winding axis in X,Y. Z=0 is the cold-core shell base plane. Form and inspect on a fixture locating that axis and the shell exits. This drawing defines the coil before final refrigeration tie-in.

2. Material: soft ACR copper, continuous and clean. No joint within the core. Form nine full turns plus the 180° upper-end clocking. The helix rises CCW from H1 to H2 when viewed from +Z.

3. Tail geometry is the tangent sequence on sheet 2. Coordinate stations are tube-centerline points. All bends are R15; preserve the stated straight lengths and exit directions. The working-tail dimensions are measured from E, before refrigeration trimming.

4. Fit to the nominal Ø127 vessel with the foil thermal interface. Over the reed bridge, raise the tube centerline 3.25 ±0.5 radially and blend smoothly at R12.7 MIN. Bridge zone Z 70.18-138.63 REF, centered on -Y. Preserve reed clearance and thermal contact elsewhere.

5. Use the undersized mandrel for rough forming; feed stock and adjust winding to the finished dimensions on the vessel fixture. Acceptance is the specified finished form, not the mandrel groove or an assumed springback.

6. Reject kinks, creases, cracks and visible flattening. Square-cut and deburr both ends; remove chips and cutting fluid, preserve the open bore and cap the ends until tie-in. Verify the formed coil and both feed-throughs before foam.

7. Refrigeration joint preparation, heat protection, brazing and leak acceptance follow assembly/refrigerant-loop.md. Trim working tails only at that assembly operation.

HSM-000015

mm DIMENSIONS GOVERN.

DO NOT SCALE.

REF = REFERENCE.

PRINT AT 100%
FOR STATED SCALES.

HOME SODA MACHINE **EVAPORATOR COIL**				**PRODUCTION PART** FORMED COIL
MATERIAL **ACR COPPER TUBE**		UNITS **MILLIMETERS**	QUANTITY **1 PER COLD CORE**	
DRAWING **EVAP-01**	REV **C**	DATE **2026-10-28**	SCALE **AS SHOWN**	SHEET **1 OF 2**

INLET TAIL / E1 - TANGENT STATIONS

FOLLOW FROM HELIX TO CUT END; ALL BENDS CL R15 ±0.5

END	ANGLE / L REF	PLANE	X	Y	Z
H1	START	-	0.000	-66.675	47.000
B0	1.72°	XZ	-0.450	-66.675	46.993
B1	50.27°	XY	-11.986	-72.088	46.993
B2	50.27°	XY	-23.522	-77.500	46.993
B3	67.60°	XZ	-37.390	-77.500	37.709
B4	67.60°	XZ	-51.258	-77.500	28.425
E1	92.24	STRAIGHT	-143.500	-77.500	28.425
CUT	200.00	STRAIGHT	-343.500	-77.500	28.425

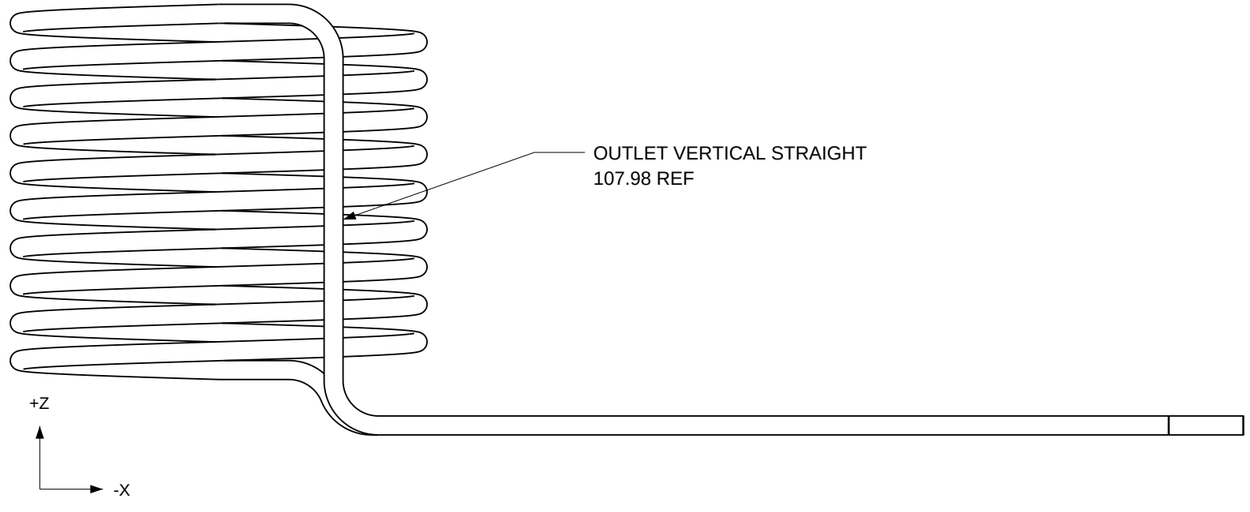
OUTLET TAIL / E2 - TANGENT STATIONS

FOLLOW FROM HELIX TO CUT END; ALL BENDS CL R15 ±0.5

END	ANGLE / L REF	PLANE	X	Y	Z
H2	START	-	0.000	66.675	166.400
B0	1.72°	XZ	-0.450	66.675	166.407
B1	50.27°	XY	-11.986	72.088	166.407
B2	50.27°	XY	-23.522	77.500	166.407
B3	90.00°	XZ	-38.522	77.500	151.407
S1	107.98	STRAIGHT	-38.522	77.500	43.425
B4	90.00°	XZ	-53.522	77.500	28.425
E2	89.98	STRAIGHT	-143.500	77.500	28.425
CUT	175.00	STRAIGHT	-318.500	77.500	28.425

SIDE - LOOKING ALONG -Y

TUBE OUTLINES - NOMINAL WINDING - SCALE 1:2



BOTH TAILS CONTINUOUS WITH THE HELIX. HIDDEN LINES OMITTED.

HOW THE FORM TABLES DEFINE THE TAILS

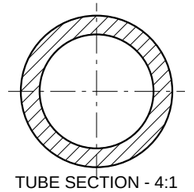
1. Each row gives the END of the identified bend or straight. Begin at H1/H2 with the helix tangent. B0 is the small pitch roll-off ending parallel to -X.

2. B1/B2 are opposite circular bends in XY: first toward the exit lane (-Y inlet, +Y outlet), then back parallel to -X. Each pair has zero intervening straight.

3. Inlet B3/B4 are opposite bends in XZ, first downward and then back parallel to -X. Outlet B3 turns downward 90°; S1 is vertical; B4 turns 90° back parallel to -X.

4. E1/E2 end the in-core straights. Continue straight along -X to CUT. Straight lengths are tangent-to-tangent; bend angles are changes in direction. Use a form fixture to reproduce the specified geometry.

5. Centerline stations: ±0.5 unless separately toleranced. Exit Y/Z: ±0.25. H2 Z and CUT coordinates are REF. Keep every junction tangent; inspect after forming and end finishing.

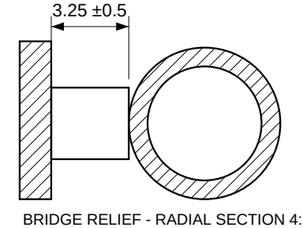


TUBE SECTION - 4:1

1/4 IN OD x 0.031 IN WALL

Ø6.35 x 0.787 WALL REF

OPEN BORE; ENDS DEBURRED



BRIDGE RELIEF - RADIAL SECTION 4:1

HSM-000015

mm DIMENSIONS GOVERN.

DO NOT SCALE.

REF = REFERENCE.

PRINT AT 100%
FOR STATED SCALES.

HOME SODA MACHINE **EVAPORATOR COIL / TAIL FORMS**				**PRODUCTION PART** FORMED COIL
MATERIAL **ACR COPPER TUBE**		UNITS **MILLIMETERS**		QUANTITY **1 PER COLD CORE**
DRAWING **EVAP-01**	REV **C**	DATE **2026-10-28**	SCALE **AS SHOWN**	SHEET **2 OF 2**